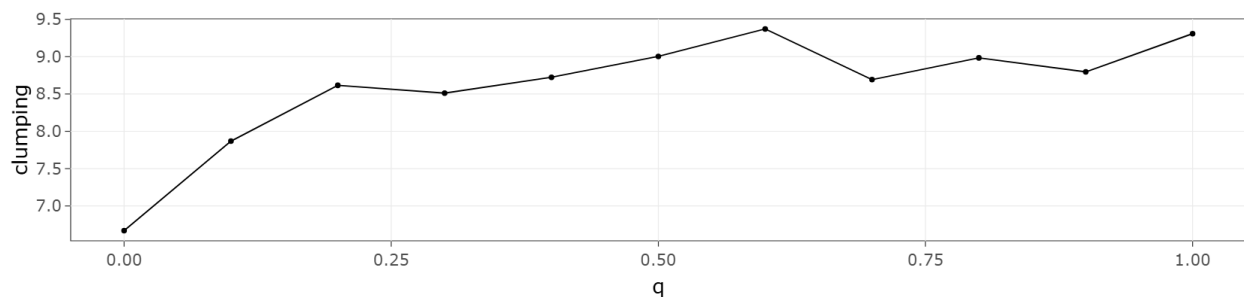


Does it seem strange that the model assumes butterflies keep moving even after finding a hilltop? Modify the model to make butterflies stop moving when no neighboring patches are higher than their current patch. You can do this by adding this statement to the beginning of the move procedure: if elevation $\geq$ max [elevation] of neighbors [stop]. First, explain exactly what each word in this statement does. Then repeat exercises 4 and 5 with this modification; does it make butterflies more or less likely to find mates? Why?

Elevation is the butterfly's current patch elevation, neighbors is the 8 surrounding patches, [elevation] of neighbors pulls their elevation values into a list, and max finds the highest one, so the whole condition is true when the butterfly's current spot is as high or higher than everything around it, meaning it's sitting on a local peak. stop then exits the move procedure for that turtle on that tick, so it just stays put. Adding this makes butterflies less likely to find mates. Without it, they keep wandering after hitting small bumps and eventually escape those bumps to converge on the one real hilltop, building up a dense clump there, but with the modification a butterfly gets permanently stuck the first time it lands on any local peak, even a tiny one, and since real terrain is full of small bumps, different butterflies end up frozen at different scattered spots instead of piling up together, which is why clumping stays low and flat across all values of q (peaking around 9) instead of climbing above 30 like it does in the unmodified version.

Modified (clumping peaks at around 9)



Unmodified (clumping climbs to about 37)

